BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 003 · Hexadecimal, shifts & masks

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Choose a type

Q2. Read an unsigned byte

Q3. Encode a negative number

Q4. Add and check the range

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I Do · Read new knowledge

Key vocabulary

Recall the term — Hexadecimal

Recall the term — Nibble

Recall the term — Logical shift

Recall the term — Mask

Recall the term — AND

Recall the term — OR

Recall the term — XOR

New learning

Hexadecimal links directly to binary — Each hexadecimal digit maps to exactly four bits because four bits have 16 possible patterns. — CODE:
$6D_{16}$
6 → 0110
D → 1101
$= 01101101_2$

Logical shifts move bits — A logical shift inserts zeros into the empty positions. Bits leaving the fixed width are lost.

AND and OR masks — AND: mask 1 preserves a bit; mask 0 clears it. — OR: mask 1 sets a bit; mask 0 preserves it.

XOR toggles selected bits — XOR produces 1 when the two input bits differ. A mask bit of 1 flips the corresponding data bit; 0 leaves it unchanged.

Worked example

Link denary, hexadecimal and binary

1. Find the hex digits — CODE:
173 ÷ 16 = 10 remainder 13 — The quotient gives the first hex digit; the remainder gives the second.

2. Convert 10 and 13 to hex — CODE:
10 = A
13 = D

$173_{10} = AD_{16}$

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3. Use four bits per hex digit — CODE:
A → 1010
D → 1101

$AD_{16} = 10101101_2$

4. Return to denary — CODE:
(10 × 16) + 13 = 173

128 + 32 + 8 + 4 + 1 = 173 — For binary → hex, group bits from the right in fours and pad the left group with zeros if needed.

We Do · Practise with guidance

2. Use hexadecimal place values

Use the method from I Do to build the answer.

Build the answer

Reverse check: 755 ÷ 16 gives remainders 3, F, 2.

3. Trace logical shifts in a fixed byte

The third shift discards the leftmost 1, so 352 cannot fit in 8 bits.

4. Preserve or clear selected bits with AND

The mask keeps the low nibble and clears the high nibble.

5. Set with OR; toggle with XOR

Divide using a shift

44 ÷ 4 = 11; the discarded bits are both zero.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Hexadecimal → binary
Write the binary equivalent of hexadecimal B7E .
Paraphrased exam question; original value and mark retained. OCR H446/01, page 17
.

Question 2

Denary → hexadecimal
Express 188_{10} in hexadecimal.
Paraphrased exam question; original value and mark retained. OCR H446/01, page 12
.

Question 3

Show how the bases connect
Convert 11100101_{2} to hexadecimal. Show the two nibble values. [2]
Convert $3A7_{16}$ to denary. Show the place-value calculation. [2]
Convert 214_{10} to hexadecimal. Show quotient and remainder or binary grouping. [2]
Original classroom extensions of the conversion skills in Questions 1–2.

Question 4

Track the stored bits
Logically shift 00110100 two places right. Give the eight-bit result and its denary value. [2]
Logically shift 10100001 one place left. Give the stored byte and explain why its

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denary value is not double the starting value. [3]
Give the stored byte after logically shifting 00001111 one place right. [1]
Adapted from OCR June 2017 Q6(c): values changed; left-shift and precision extensions added. Classroom marks.

Question 5

Apply each operation independently
Start with 10100101 each time. Give the result of:
AND with 00001111 . [1]
OR with 01000010 . [1]
XOR with 11110000 . [1]
Original exam-style practice aligned to PMT bitwise manipulation and masks.

Question 6

Control an eight-bit register
Bits are numbered 7…0 from left to right. Give an operation and an eight-bit mask for each independent task. Preserve all bits not mentioned.
Clear bits 7 and 6. [2]
Set bits 3 and 0. [2]
Toggle bits 5 and 2. [2]
Original exam-style application; one mark for the operation and one for the mask.

Exit Ticket · Answer before checking

One hexadecimal digit represents four binary bits.

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Boolean: True or False.

Q2. 64 + 16 + 8 + 2 = 90 .

Q3. +18 = 00010010 . Sign magnitude: 10010010 . Two’s complement: invert → 11101101 , add 1 → 11101110 .

Q4. 10010000 . 112 + 32 = 144, outside −128…127. Signed overflow occurs; the stored pattern represents −112 when interpreted as an 8-bit two’s-complement integer.

You Do pages · Independent questions

Q1. B = 1011; 7 = 0111; E = 1110. Answer: 1011 0111 1110 (1). Retain four bits per hexadecimal digit.

Q2. 188 ÷ 16 = 11 remainder 12 → BC (1). Check: 11 × 16 + 12 = 188.

Q3. 1110 = E and 0101 = 5 (1); E5 (1).
(3 × 256) + (10 × 16) + 7 (1) = 935 (1).
214 ÷ 16 = 13 remainder 6 (1); D6 (1).

Q4. 00001101 (1), 13 (1).
01000010 (1). The leftmost 1 is shifted out and discarded (1); 161 × 2 = 322 exceeds 255, while the stored value is 66 (1).
00000111 (1). The stored value is 7, not 7.5.

Q5. CODE:
10100101 10100101 10100101
AND 00001111 OR 01000010 XOR 11110000
---------- ---------- ----------
00000101 11100111 01010101
One mark per correct result. Each operation starts from the original byte.

Q6. AND (1) with 00111111 (1). Zeros clear bits 7 and 6; ones preserve the rest.
OR (1) with 00001001 (1). Ones set bits 3 and 0.
XOR (1) with 00100100 (1). Ones toggle bits 5 and 2.

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Q1–Q2 retain the original exam mark allocations. Q3–Q6 use classroom allocations. Answers are calculated teaching guidance, not a reproduced official OCR mark scheme.

Exit Ticket · Model answer

True. Each hexadecimal digit represents four binary bits.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: